

**PRAGATI ENGINEERING COLLEGE: SURAMPALEM
(AUTONOMOUS)**

II B.Tech II Semester Regular Examinations, April - 2025

**OPERATING SYSTEMS
(Common to CSE, IT and CSE(CS))**

Time: 3 hours

Max. Marks: 70

Note:

- i. Question No. 1 shall contain 10 compulsory short answer questions (2 questions from each unit) for a total of 20 marks such that each question carries 2 marks.
- ii. In each of the questions from 2 to the last question, there shall be either/or type questions of 10 marks each.

Q. No.	Questions	BTL	CO	Marks
1.	a) State any two memory management functions of an OS.	K1	CO1	2M
	b) Define bootstrap program.	K1	CO1	2M
	c) What are the different states of a process?	K1	CO2	2M
	d) Name any two CPU scheduling algorithms.	K1	CO2	2M
	e) What is a critical section in operating systems?	K1	CO3	2M
	f) How does a wait-for graph help in detecting deadlocks?	K1	CO3	2M
	g) What is a page table?	K1	CO4	2M
	h) What is FCFS in HDD scheduling?	K1	CO4	2M
	i) Name the three types of file access methods.	K1	CO5	2M
	j) How does domain define access rights?	K1	CO5	2M
UNIT – I				
2.	a) Describe the characteristics and licensing of open-source operating systems.	K2	CO1	5M
	b) How does the OS provide services for program execution and I/O operations?	K2	CO1	5M
OR				
3.	a) List and explain different types of system calls with examples.	K2	CO1	5M
	b) Discuss the role of user goals and system goals in OS design.	K2	CO1	5M
UNIT – II				
4.	a) Define a thread. Discuss the benefits of multithreading.	K1	CO2	5M
	b) Compare and contrast the different multithreading models.	K2	CO2	5M
OR				
5.	a) What are thread libraries? Describe any two thread libraries with features.	K2	CO2	5M
	b) Explain the different types of thread cancellation.	K2	CO2	5M
UNIT – III				
6.	a) What is a mutex lock? Explain how it can be used to solve the critical section problem.	K1	CO3	5M
	b) Describe the Readers-Writers problem with appropriate synchronization mechanisms.	K3	CO3	5M

OR

7.	a)	List and explain the four general strategies to handle deadlocks.	K2	CO3	5M
	b)	Describe Banker's Algorithm with a step-by-step example.	K3	CO3	5M

UNIT – IV

8.	a)	Discuss different types of contiguous memory allocation techniques.	K2	CO4	5M
	b)	How does swapping affect system performance and memory utilization?	K1	CO4	5M

OR

9.	a)	Explain the FIFO and LRU page replacement algorithms with examples.	K2	CO4	5M
	b)	Compare and contrast various page replacement strategies.	K2	CO4	5M

UNIT – V

10.	a)	Discuss how indexed allocation overcomes the drawbacks of linked allocation.	K2	CO5	5M
	b)	How is a free space list maintained and used during file allocation?	K1	CO5	5M

OR

11.	a)	What is mounting? Explain how file systems are mounted and unmounted.	K2	CO5	5M
	b)	Compare static and dynamic partitioning methods.	K2	CO5	5M